

VOLTIX

COMPLETE CONTRACT & RULE SPECIFICATION

Version 1.1 | Architectural and Semantic Baseline

Document intent This document defines the semantic contracts, decision authority, rule boundaries, state meanings, failure behavior, and implementation obligations for VOLTIX V1. It is intentionally not a code, API, SQL, firmware, or technology-stack specification.

Field	Value
Product	VOLTIX
Program	SIH26219 / Smart Automation / Hardware
Document type	Contract Specification (CS) - architecture + semantics
Version	1.0
Status	Baseline for implementation and review
Primary audience	Platform, edge, rules, ML, hardware, safety, QA, M&V, and application developers
Source basis	VOLTIX final detailed technical specification; VOLTIX final quick-reference document; Rule Engine Ticket
Scope	Ingest/DQ through Sense -> Verify -> Decide -> Act -> Prove

How to use this specification

1. Treat semantic meaning, authority boundaries, safety behavior, and observable outcomes as locked V1 contracts unless an explicit decision changes them.
2. Treat programming language, framework, database, message broker, deployment packaging, and internal class/module structure as implementation freedom.
3. Implementers must preserve canonical upstream outputs and must not silently recreate upstream intelligence in downstream layers.
4. When a rule is marked policy- or parameter-driven, do not hard-code a universal numeric value without a registered basis and lifecycle.
5. When this document conflicts with a source document, record the discrepancy and resolve it as an explicit architecture decision; do not silently reconcile it.

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PART I | DOCUMENT CONTROL & ARCHITECTURE

Purpose. Establishes what the CS governs, what is locked, what remains implementation-free, and how the VOLTIX architecture is partitioned.

I.1 Source-of-truth hierarchy

Priority	Source	Use
1	VOLTIX final detailed technical specification	Primary authority for architecture, product boundaries, safety, M&V, research basis, hardware/software responsibilities and honest limits.
2	VOLTIX final quick-reference document	Operational reference for concise V1

Priority	Source	Use
		interpretation, product boundaries and current-stage statements.
3	Rule Engine Ticket	Formal source for Ranker, Alerts and adjacent product-rule ownership/boundaries; no ML ownership.
4	This CS	Formal semantic and cross-layer contract baseline synthesized from the above. Where this CS makes a new architectural constraint, it is an explicit contract decision, not a silent correction of the sources.

I.2 Product contract

VOLTIX is a low-cost retrofit measurement, decision and proof layer for legacy MSME factories. V1 continuously observes per-machine electrical behaviour, interprets operating state, quantifies avoidable energy cost using estimated power where necessary, ranks losses, raises explainable alerts, optionally actuates only approved safe loads, detects ACTIVE operating drift, and exports operational M&V-style evidence.

Locked product loop SENSE -> VERIFY -> DECIDE -> ACT -> PROVE

I.3 Architecture doctrine

Doctrine	Contract meaning
Brownfield first	The system must tolerate legacy machines, imperfect connectivity, limited instrumentation and installation constraints.
Safety over cleverness	Ambiguity must bias toward no actuation and preserved production continuity.
Evidence before authority	Upstream layers provide evidence; decision/policy layers determine authority; actuation remains locally fail-safe.
Explainable decisions	Every material recommendation or action must have a machine-readable reason code and human-readable explanation.
Proof is product	Savings claims require baseline/reporting context, assumptions, normalization and verification status.
Condition is first-class	Condition/drift is a peer product capability, not a hidden annotation inside waste detection.
Honest physics	Estimated CT-only power is useful for operations but is not represented as billing-grade measurement.
Pareto installs	V1 favours one primary CT per prioritized machine rather than requiring whole-facility intrusive instrumentation.

I.4 Semantic lock vs implementation freedom

Locked semantic contract	Implementation freedom
State names, meanings, evidence statuses and authority	Programming language, framework and internal object model

Locked semantic contract	Implementation freedom
hierarchy	
Required inputs/outputs and provenance	API transport format and serialization technology
Fail-safe behaviour, local safety authority and AutoCut deny rules	Edge OS, database, queue and broker selection
Rule lifecycle and version semantics	Repository layout and deployment pipeline
Audit and decision-trace requirements	Storage schema and query strategy
M&V meaning and evidence status	PDF/CSV generation library and reporting UI implementation

I.5 V1 scope boundaries

- CT-first per-machine observation; V1 estimated power may use nominal voltage and an assumed PF.
- Pulse state model with OFF, ACTIVE, IDLE and WASTE, plus separate classification status for uncertainty or unavailability.
- Energy residual for avoidable power/cost while state and baseline evidence are valid.
- ACTIVE-only electrical drift detection, primarily as condition awareness rather than full predictive maintenance.
- Explainable ranker and alerting rules; no black-box rule weights.
- Policy-gated AutoCut only for explicitly eligible and whitelisted loads; cloud loss never forces a cut.
- Operational M&V-style reporting with explicit estimation/verification status.

I.6 Non-goals / honest limits

- Utility billing accuracy.
- Full predictive-maintenance or failure-day prediction.
- Good-part or quality-yield counting from current alone.
- Air-leak localization without flow sensing.
- Unrestricted shutdown of arbitrary production machinery.

PART II | CONTRACT STANDARDS & SHARED PRIMITIVES

Purpose. Defines shared identity, time, versioning, evidence, unknown semantics, windowing, decision and audit primitives used across every layer.

II.1 Entity and identity model

Entity	Semantic responsibility	Identity rule
Site	Factory or operational boundary	Globally unique site identity; all downstream evidence must resolve to a site.
Machine	Tracked load/asset	Stable machine identity; one primary CT node maps to one machine in V1.
Node / Device	Installed sensing/actuation endpoint	Stable device identity plus boot/session identity for replay and reboot reasoning.
Gateway	Local aggregation / edge authority	Stable gateway identity; participates in local availability and synchronization state.
Telemetry point	Timestamped observed measurement	Identity combines device, machine, event

Entity	Semantic responsibility	Identity rule
		time and sequence/provenance as applicable.
Pulse state event	Canonical operating-state transition or observation	Machine-scoped; references Pulse model/rule provenance.
Residual estimate	Energy/economic interpretation	Machine + time-window scoped; carries baseline and measurement provenance.
Condition score	ACTIVE-only drift interpretation	Machine + operating regime + model/version scoped.
Rank snapshot	Owner-facing priority view	Site-scoped snapshot with ranking provenance and freshness.
Alert	Stateful notification lifecycle	Machine + alert-type scope; dedupe and lifecycle state required.
Action command	Requested actuation	Targeted, authenticated, expiring, idempotent command; never equivalent to physical outcome.
M&V baseline/report	Savings evidence	Site/machine/project context with explicit period and assumptions.
Policy / parameter	Authority or configurable threshold	Versioned, effective-dated and provenance-bearing.

II.2 Canonical time model

Field	Meaning	Contract
event_time	When the measurement/state/action actually occurred	Primary temporal reference for engineering meaning.
ingest_time	When the platform accepted the record	Used for freshness, delay and transport diagnostics.
processing_time	When an engine evaluated the record	Used for system latency and replay analysis.
effective_time	When a policy/parameter becomes active	Must be explicit; never infer from publication time.
observed_age	Current time - event_time	Must be available to freshness-sensitive consumers.

Time rule Event time is the semantic clock. Processing or arrival time must not silently replace event time when determining state duration, residual windows, alerts or M&V periods.

II.3 Version taxonomy

Version type	What changes it	Examples
contract_version	Field meanings, required statuses, structural semantics	CS v1.0
rule_version	Decision logic	rules-ranker-v1; rules-alerts-v1

Version type	What changes it	Examples
model_version	Learned/statistical model artefact	gmm-v1; IF-v1
policy_version	Authority/allow-deny configuration	site-policy-v3
parameter_version	Threshold/calibration values	energy-params-v2
baseline_version	M&V baseline definition	baseline-machine-17-v4
firmware_version	Node runtime implementation	device firmware identifier
configuration_version	Deployment/binding configuration	commissioning config version

Version rule Changing a threshold is not the same semantic event as changing a rule. Every decision must retain enough provenance to identify both.

II.4 Evidence and data-quality status

Status	Meaning	Downstream implication
VALID	Evidence meets freshness, completeness and semantic checks	May be consumed subject to layer-specific gates.
STALE	Evidence exists but is outside the freshness contract	May support diagnosis/history; must not silently drive time-sensitive action.
ESTIMATED	Derived using assumptions or proxies	Allowed for operations; claims must preserve estimation provenance.
VERIFIED	Evidence has passed the applicable verification procedure	Can support verified M&V/operational claim where scope matches.
UNCERTAIN	Evidence exists but classification/confidence is ambiguous	May support human review; safety-relevant automatic action is blocked.
UNAVAILABLE	Required evidence not present	Decision must degrade explicitly; never fabricate a default.
INVALID	Evidence fails a hard contract or integrity check	Quarantine/ignore for decisioning until recovery.

II.5 Unknown semantics

UNKNOWN, UNCERTAIN, FALSE and UNAVAILABLE are distinct. An unknown context value does not equal a negative fact. A missing schedule does not mean the machine is outside schedule. A low-confidence Pulse classification does not equal IDLE.

II.6 Window primitives

Primitive	Meaning	Required semantics
point	Single timestamped observation	No implied averaging.
rolling window	Moving time window	Window anchor and inclusion rule must be explicit.
state duration	Elapsed time since current canonical state became effective	Derived from state_since and event_time.
persistence window	Minimum continuous duration for	Must account for gaps and stale evidence.

Primitive	Meaning	Required semantics
	promotion/trigger	
baseline window	Period used to define necessary/expected behaviour	Versioned; must have validity status.
reporting window	Period over which evidence is summarized	Must be comparable with the baseline scope.

II.7 Shared decision primitive

Every material decision is represented semantically as: decision outcome + rule_id + rule_version + reason_code + explanation + evaluated_at + context timestamp + provenance. Safety actions additionally require authorization/policy context, command identity and eventual physical-outcome verification.

II.8 Rule type taxonomy

Rule type	Question answered	Examples
Interpretation	What does the evidence mean?	Pulse state classification
Validation	Can this evidence be trusted?	Telemetry range/sequence checks
Quantification	How much is happening?	Avoidable kW / INR per hour
Qualification	Is the situation worth considering?	Sustained-waste eligibility
Decision	What should the product do?	Rank, alert, dispatch recommendation
Policy	What is authorized?	Whitelist / deny / approval state
Temporal	Does timing change the result?	On-delay, cooldown, persistence
Safety	What must never happen?	Fail-safe ON, no AutoCut on never-AUTO load
Presentation	How should meaning be shown?	Estimated/stale labels, explanation copy
Proof	Can a claim be substantiated?	M&V verification state

II.9 Canonical rule evaluation context

Every rule-bearing engine evaluates against a coherent Rule Evaluation Context rather than independently rebuilding evidence. The context is a logical snapshot, not necessarily a single database object. Its purpose is semantic consistency and traceability.

Context domain	Required semantic content	Authority meaning
Identity	site_id, machine_id, device_id, asset class, binding status, tenancy scope	Identity establishes scope; it does not authorize action.
Telemetry	latest eligible measurements plus event/ingest timestamps and provenance	Observation only.
Data Quality	valid/stale/invalid/unavailable state, completeness, sequence/replay state	Determines whether evidence may be trusted.
Pulse	canonical state, classification_status, confidence, state_since, model/rule provenance	Interpretation of operating state; no action authority.

Context domain	Required semantic content	Authority meaning
Energy	estimated power, baseline reference, residual, cost rate, validity/provenance	Quantification; no shutdown authority.
Condition	operating regime, drift score/flag, model readiness, persistence state	Condition awareness; no shutdown authority.
Temporal	event time, duration, schedule, shift, maintenance/planned downtime, freshness age	Timing/context only.
Policy	mode, whitelist, deny/allow decisions, approvals, expiry, override state	Authority and permission.
Hardware capability	actuator presence/rating, interlocks, feedback capability, device health	Physical feasibility and safety gating.
Existing state	open alerts, prior action state, lockout state, acknowledgement/suppression state	Stateful continuity.
Version provenance	contract/rule/model/policy/parameter/baseline/config/firmware versions	Reproducibility and audit.

Context consistency rule Downstream engines must consume canonical upstream outputs. Reimplementation of Pulse, Energy or Condition semantics inside Ranker, Alerts, Action or Apps is prohibited unless a new rule explicitly owns the changed interpretation and updates the contract.

II.10 Separate status taxonomies

VOLTIX must not collapse freshness, provenance, classification quality and decision state into a single status. These are orthogonal dimensions.

Dimension	Controlled values	Meaning
Data quality status	VALID / STALE / INVALID / UNAVAILABLE	Can this evidence be trusted for the relevant contract?
Evidence provenance	MEASURED / CALCULATED / ESTIMATED / INFERRED / VERIFIED	How was this value produced and, where applicable, verified?
Classification status	VALID / UNCERTAIN / UNAVAILABLE	Can the classifier support a canonical interpretation?
Decision status	CANDIDATE / QUALIFIED / ELIGIBLE / INELIGIBLE / BLOCKED / DENIED / SUPPRESSED / DEFERRED / EXECUTED / VERIFIED / FAILED	What semantic decision state does the rule evaluation occupy?
Claim status	ESTIMATED / VERIFIED / INVALID	What strength of external/operational claim is supported?

II.11 Universal units and numeric semantics

Rule	Contract
Unit declaration	Every numeric quantity must have an explicit unit or dimensionless declaration.
Range declaration	Each contract field defines allowed physical/logical range,

Rule	Contract
	including equality boundaries.
Precision	Storage precision may exceed display precision; calculation precision must not be silently reduced for convenience.
Rounding	Rounding is a presentation concern unless a rule explicitly states that rounded values are decision inputs.
Sign convention	Power/cost/residual signs must be defined once; avoidable waste is non-negative by contract.
Nullability	Missing, unknown, not applicable and unavailable are distinct semantics.
NaN/infinity	Non-finite numeric values are invalid inputs, never silently coerced to zero.

II.12 Shared decision-state semantics

Decision state	Meaning	Typical transition
CANDIDATE	Evidence suggests a situation may qualify for a rule.	Observed -> candidate
QUALIFIED	All qualification conditions passed, but authority may still be missing.	Candidate -> qualified
ELIGIBLE	Qualification plus all required non-authority gates passed.	Qualified -> eligible
INELIGIBLE	Known evidence does not satisfy required conditions.	Candidate/qualified -> ineligible
BLOCKED	Action/decision is prevented by a higher-priority safety, policy or integrity gate.	Eligible -> blocked
DENIED	An explicit policy/authority decision prohibits the requested outcome.	Eligible -> denied
SUPPRESSED	Outcome is intentionally withheld by temporary lifecycle/context policy.	Qualified -> suppressed
DEFERRED	Outcome requires later evidence or timing before evaluation can complete.	Candidate -> deferred
EXECUTED	Requested action was accepted by the execution path.	Authorized -> executed
VERIFIED	Expected result has been independently verified within the contract.	Executed -> verified
FAILED	Execution or required verification failed.	Executed -> failed

II.13 Reusable rule specification template

Every rule in the registry must be expressible using the following semantic template so implementation teams do not infer missing behaviour.

Field	Required meaning
Rule ID / name	Stable identifier and concise semantic name.
Rule class	Interpretation / Validation / Quantification / Qualification / Decision / Policy / Temporal / Safety / Presentation / Proof.
Authority class	Physical / Statistical / Safety / Policy / Site / Machine / Product / Business / Configuration as applicable.
Owner	Responsible domain/product owner and approval role.
Purpose	One sentence describing the business/system question answered.
Inputs	Canonical upstream fields and status dimensions.
Preconditions	Required evidence, freshness, binding, readiness and context conditions.
Logic	Formal condition, formula, transition or decision table stated in natural-language semantics.
Parameters	Parameter IDs, units, provenance and effective version.
Temporal semantics	Persistence, delays, windows, expiry and ordering rules.
Conflict priority	What happens if another rule disagrees; identify higher authority.
Output	Canonical fields, decision state, reason code and human explanation.
Failure/degraded behaviour	Required result when evidence is missing, stale, uncertain or invalid.
Audit requirement	What must be recorded to reconstruct the evaluation.
Test obligation	Acceptance, boundary, failure and adversarial cases.
Version/provenance	Rule/version plus supporting source or decision basis.

PART III | LAYER 0 - INGEST / DATA QUALITY RULES

Purpose. Defines the trustworthy evidence gate between raw device traffic and all downstream engines.

III.1 Responsibilities

- Device authentication and identity resolution
- Schema and semantic validation
- Timestamp and clock-quality handling
- Machine binding and tenancy/site isolation
- Deduplication and replay detection
- Out-of-order and sequence handling
- Completeness/freshness/rate health metrics
- Buffering and offline continuity
- Quarantine and recovery
- Commissioning completeness state

III.2 Canonical telemetry minimum

Field	Meaning	Contract
device_id	Node identity	Required; authenticated.
site_id	Site binding	Required; must agree with configuration.
machine_id	Machine binding	Required after commissioning; unbound telemetry cannot train Pulse.
ts	Event timestamp	Required; checked for plausibility and monotonicity where applicable.
i_rms_a	RMS current	Required for CT-first power estimation.
v_nominal	Nominal supply voltage	Configuration/provenance required; not treated as a real-time measurement unless instrumented.
pf_assumed	Power factor assumption	Explicitly flagged as assumed.
kw_est	Derived estimated power	Derived value; provenance must identify formula and assumptions.
temp_c	Optional enclosure/device temperature	Quality and condition context only unless explicitly bound to a contract.
link_quality	Connectivity health	Diagnostics; not machine condition.
firmware_version	Node software provenance	Required for device traceability.
boot_id	Boot/session identity	Supports reboot, replay and continuity reasoning.

III.3 DQ rule catalogue

Rule ID	Rule	Outcome / action
DQ-001	Authenticated device identity required	Reject/quarantine unauthenticated data.
DQ-002	Known site and machine binding required for decision use	Accept to raw store if allowed; exclude from machine intelligence if unbound.
DQ-003	Timestamp plausibility	Mark INVALID/STALE or quarantine according to severity.
DQ-004	Duplicate/replay detection	Deduplicate without re-triggering downstream state/action.
DQ-005	Sequence continuity	Record gap; do not fabricate missing measurements.
DQ-006	Out-of-order tolerance	Late data may enter history but cannot silently rewrite already-executed safety decisions.
DQ-007	Range sanity	Reject impossible values; flag suspicious but physically possible values.
DQ-008	Rate/flood protection	Throttle/quarantine abusive or malfunctioning streams.
DQ-009	Freshness contract	Publish explicit freshness state to

Rule ID	Rule	Outcome / action
		downstream consumers.
DQ-010	Buffer integrity	Preserve event time/order metadata across offline buffering.
DQ-011	Commissioning completeness	Machine remains commissioning-incomplete until required capture/health checks pass.
DQ-012	Recovery re-admission	Quarantined streams require successful validation before re-entering decision path.

III.4 Offline behaviour

During gateway/cloud loss, nodes/gateways buffer telemetry within designed capacity, preserve event timestamps and continue safe local operation. Cloud unavailability must never be interpreted as authorization to cut. Upon reconnection, late telemetry is reconciled as historical evidence; it must not retroactively create a command or duplicate an action already decided.

III.4 Commissioning lifecycle

Lifecycle state	Meaning	Exit requirement
UNCOMMISSIONED	Asset exists in inventory but installation/binding contract is incomplete.	Installation and identity prerequisites recorded.
INSTALLATION_PENDING	Physical installation/configuration work is incomplete.	CT/actuator/node installation checks complete.
BOUND	Node is bound to the intended site/machine.	Binding validated and identity consistent.
CAPTURE_IN_PROGRESS	Required OFF/ACTIVE/IDLE evidence is being collected.	Required capture set complete and quality-valid.
VALIDATION_PENDING	Capture, tariff, hardware, health and safety checks are being reviewed.	All mandatory commissioning rules pass.
COMMISSIONED	Machine is admitted to normal system operation under its declared capabilities and policy.	Automatic revalidation remains ongoing.
OPERATIONAL	Commissioned machine is healthy enough for normal decision use.	Continues while health/freshness/policy remain valid.
SUSPENDED	Previously commissioned machine is temporarily excluded from decision/action due to integrity, maintenance or policy conditions.	Explicit requalification and required checks pass.

III.5 Commissioning rule catalogue

Rule ID	Rule	Semantic result
COM-001	Identity and site binding validation	Machine/device relationship must be explicit and unique for decision use.
COM-002	CT installation/conductor validation	CT placement and polarity/association

Rule ID	Rule	Semantic result
		must be verified before using current as machine evidence.
COM-003	OFF capture validation	A representative OFF condition must show expected near-zero/negligible draw for the installed sensing path.
COM-004	ACTIVE capture validation	A representative productive/loaded condition must be captured and quality-checked.
COM-005	IDLE capture validation	A representative powered/nonproductive condition must be captured where the machine supports it.
COM-006	Tariff/price provenance validation	Economic rate source and effective period must be known for monetary outputs.
COM-007	Node/gateway health validation	Heartbeat, buffering and connectivity behaviour must pass the declared deployment contract.
COM-008	Override validation	Manual override path must be demonstrated and its recovery/latch semantics recorded before AutoCut eligibility.
COM-009	Actuator and interlock validation	Installed hardware must be correctly mapped, rated, and physically compatible with the approved action path.
COM-010	Never-AUTO classification validation	Machine safety classification and explicit AutoCut whitelist status must be recorded.
COM-011	Baseline readiness validation	Energy baseline must meet its validity/readiness contract before decision-grade residual claims.
COM-012	Model readiness validation	Any model used in production must have required lifecycle state and version provenance; otherwise rules-v1 or a degraded status applies.

III.6 Commissioning suspension rules

Loss of a required commissioning invariant must not silently leave a machine in its prior privileged state. Examples include binding mismatch, CT replacement, actuator replacement, failed override test, policy revocation, hardware rating change or material process reconfiguration. The system may continue observation where safe, but privileged decisions require requalification.

PART IV | LAYER 1 - PULSE CONTRACT / RULE BOUNDARY

Purpose. Defines the canonical machine operating-state contract and the boundary between rule-based state inference and later statistical models.

IV.1 Canonical states

State	Meaning	Important distinction
OFF	Negligible draw / effectively off	Not the same as telemetry unavailable.
ACTIVE	Productive or loaded operation	Condition/drift logic is primarily evaluated here.
IDLE	Powered but nonproductive / unloaded behaviour	May become WASTE after policy/time/context criteria.
WASTE	Policy-labelled avoidable/prolonged nonproductive powered behaviour	WASTE is not simply “high current” or “IDLE forever”.

IV.2 Classification status

classification_status	Meaning	Actionability
VALID	Canonical state confidently determined	Normal downstream use subject to freshness and policy.
UNCERTAIN	Evidence ambiguous / low confidence	History/UI may show state; AutoCut blocked.
UNAVAILABLE	No usable classification	Downstream qualification requiring Pulse must fail closed.

IV.3 Pulse contract

Output	Required meaning
state	OFF/ACTIVE/IDLE/WASTE only
classification_status	VALID/UNCERTAIN/UNAVAILABLE
state_confidence	Numerical/ordinal confidence according to model contract; not a universal AutoCut threshold
state_since	Event time from which current state is considered active
model_or_rule_version	Provenance of classification logic
evidence_freshness	Current/STALE/UNAVAILABLE as applicable

IV.4 Rules-v1 and GMM boundary

V1 may operate with deterministic commissioning thresholds and later introduce GMM-v1. The GMM is an implementation of classification intelligence, not a redefinition of downstream semantics. Low maximum GMM responsibility, high entropy or otherwise insufficient evidence results in holding the previously debounced state or marking classification uncertain according to the model contract. Downstream layers consume the canonical state and must not retrain or reclassify it.

IV.5 Debounce and transition semantics

- Transient threshold crossings must not immediately become state transitions.
- State promotion/demotion uses configured temporal persistence/hysteresis.
- state_since resets only when a new state becomes effective under the debounce contract.
- A telemetry gap cannot be silently interpreted as a state transition.

- Transition provenance must retain both candidate evidence and final accepted state.

PART V | LAYER 2 - ENERGY RULES

Purpose. Defines estimated power, baselines, residual waste and economic quantification while preserving honest measurement provenance.

V.1 Power estimation

For CT-first V1, estimated power may be represented as $kw_est = (i_rms_a * v_nominal * pf_assumed) / 1000$. The formula is an estimate because voltage and power factor may be assumptions rather than synchronized measurements. For properly metered balanced three-phase systems, the semantic reference may instead be $P = \sqrt{3} * V_{LL} * I * PF$. The implementation must retain the provenance of which method was used.

V.2 Residual contract

While Pulse indicates a nonproductive or qualifying state, avoidable power is conceptually $P_{avoidable} = \max(0, P_{estimated} - P_{baseline_necessary})$. The residual must never be presented without the baseline-validity state that makes the comparison meaningful.

V.3 Baseline hierarchy

Baseline type	Use	Validity conditions
Commissioning idle floor	Initial reference for powered nonproductive behaviour	Derived from captured machine-specific commissioning evidence.
Rolling statistical idle baseline	Adaptive reference as operation evolves	Requires clean, comparable samples and contamination controls.
Process-aware necessary-power baseline	Best available representation of required power	Requires operating regime/process context and explicit calibration.

V.4 Energy outputs

Output	Meaning
waste_kw	Current avoidable power estimate
waste_kwh_today	Accumulated avoidable energy estimate over reporting day/window
waste_inr_today	Economic estimate using applicable tariff
waste_inr_per_hr	Current avoidable cost rate
baseline_validity	VALID/STALE/INVALID/UNAVAILABLE and reason
measurement_status	ESTIMATED/VERIFIED as applicable
tariff_version	Economic-rate provenance

V.5 Quantification rules

Rule ID	Rule	Semantic outcome
ENG-001	No valid Pulse evidence -> no state-conditioned residual claim	Return unavailable/not-qualified rather than guess.
ENG-002	Invalid baseline -> residual not decision-	Display degraded status and exclude from

Rule ID	Rule	Semantic outcome
	grade	ranking/AutoCut eligibility where required.
ENG-003	Negative residual -> clamp avoidable portion to zero	Do not report negative waste.
ENG-004	Tariff missing	Retain physical residual if valid; economic values become unavailable/estimated rather than zero.
ENG-005	Assumed PF/nominal V	All economic/energy outputs inherit ESTIMATED provenance.
ENG-006	Accumulation across outages	Use valid event-time intervals only; never fill gaps with fabricated energy.

V.6 Honest interpretation

Product truth V1 energy numbers are decision-support estimates, not utility-billing measurements. Any UI or M&V export must preserve the assumptions behind the estimate.

PART VI | LAYER 3 - CONDITION RULES

Purpose. Defines ACTIVE-only electrical drift intelligence and prevents condition analytics from becoming unrestricted predictive maintenance or actuation authority.

VI.1 Scope

Condition is a first-class capability operating primarily during ACTIVE periods. It is intended to flag electrical behaviour that drifts from a healthy reference for the same machine/operating regime. It is not a guarantee of impending failure and does not authorize shutdown of CNCs, furnaces or other never-AUTO loads.

VI.2 Feature family

Feature	Semantic use
ACTIVE RMS mean	Central operating-current level
spread / variance / IQR	Stability and variability
short-term trend slope	Direction of change
distance from ACTIVE centroid	Deviation from learned healthy operating signature
optional enclosure temperature	Supporting device/machine context where valid

VI.3 Training eligibility

- Training/reference data must primarily represent known normal operation.
- Pulse must indicate ACTIVE with valid classification.
- Data-quality gates must pass freshness/range/completeness checks.
- Known abnormal periods, maintenance interventions, commissioning anomalies and contaminated samples must not silently become the healthy baseline.
- Operating-regime separation should be used where one machine has materially different normal modes.

VI.4 IF model and threshold calibration

An Isolation Forest or similar anomaly model may be used, but its score is not universally interpretable as “50% abnormal”. Production thresholds must be calibrated from healthy validation data and operational review. Literature examples such as contamination fractions or high-percentile thresholds are candidate starting points, not VOLTIX universal constants.

VI.5 Condition contract

Output	Contract
drift_score	0-1 or standardized anomaly score, with model-specific meaning recorded
flag	NONE / ADVISORY / ACTIONABLE-REVIEW according to policy
severity	Product severity, distinct from raw anomaly score
message	Explainable reason for the flag
model_version	Condition model provenance
operating_regime	Context under which comparison is valid
active_only_guarantee	Explicit indication that score was evaluated from ACTIVE-qualified evidence
baseline_or_model_status	READY / SHADOW / STALE / INVALID / UNAVAILABLE

VI.6 Persistence and reset

- A single anomalous sample should not normally create a durable condition alert.
- Persistence, deadband and cooldown are policy/parameter contracts, not hard-coded science.
- Major maintenance or process changes may invalidate a previous healthy reference and trigger re-baselining.
- Condition drift must not change Pulse state, energy baseline or action policy by side effect.

PART VII | LAYER 4 - WASTE RANKER

Purpose. Produces a stable, explainable owner-priority list from canonical upstream evidence; it does not reclassify machine states or hide policy decisions.

VII.1 Ranker objective

Answer: “Which machine should the owner look at or fix first right now?” The V1 ranker should prioritize current qualified avoidable cost rate rather than an opaque arbitrary weighted score. Accumulated cost, Pareto contribution, criticality and condition are secondary views/annotations.

VII.2 Three distinct concepts

Concept	Meaning	Do not conflate with
PRIORITY	What deserves attention first now	Long-term significance
SIGNIFICANCE	What has accumulated/material historical cost	Immediate action urgency
AUTOCUT_OPPORTUNITY	What may be eligible for safe automatic action	Human priority ranking

VII.3 Primary eligibility

Gate	Must be true
Identity	Known site + machine
Freshness	Pulse and energy evidence within decision freshness contract
Pulse	Canonical nonproductive state or another explicitly qualifying condition
Duration	Qualifying condition persisted beyond minimum threshold
Baseline	Energy baseline valid for current comparison
Quantification	Avoidable cost rate available or explicit degraded status
Policy	No policy denial for presentation/decision use
Safety	Any action annotation must independently pass action policy; ranking itself does not authorize action

VII.4 Primary ordering

Primary ordering: higher qualified waste_inr_per_hr ranks first. Tie-breakers should prefer longer qualifying duration, then larger waste_kw, then explicit machine criticality, then stable machine identity to avoid churn. Ranking must be stable within the configured hysteresis unless material evidence changes justify a reorder.

VII.5 Secondary scoring/annotation

- Show current INR/hr prominently.
- Show current waste duration.
- Show accumulated INR separately.
- Show drift condition as an annotation, not a multiplier that hides cost.
- Show AutoCut eligibility as a separate badge with deny reason when not eligible.
- Where Pareto is displayed, keep cumulative share as a view over ranked cost contribution rather than replacing immediate priority.

VII.6 Ranker contract

Output	Meaning
rank	Ordinal position in snapshot
score	Optional normalized internal ranking value; must be explainable
primary_metric	The metric actually used for order
explain	Human-readable one-sentence reason
reason_code	Machine-readable classification of why the machine is ranked
suggested_action	Human recommendation; never a command
criticality	Optional machine/site context
condition_annotation	Optional drift status
autocut_annotation	Eligible / denied / not assessed with reason
rule_version	Ranker rule provenance
snapshot_time	Evaluation timestamp

VII.7 Anti-patterns

- Do not multiply cost by confidence into a “fake” score that obscures the actual rupee impact.
- Do not rank by raw current alone.
- Do not treat a drift score as monetary waste.
- Do not rank stale/invalid evidence as if it were current.
- Do not let frontend code invent its own ordering formula.

PART VIII | LAYER 5 - ALERTS

Purpose. Defines stateful alert lifecycles, anti-spam controls, deduplication, escalation and the boundary between notifying a human and authorizing an action.

VIII.1 Alert lifecycle

State	Meaning
CREATED	Condition first evaluated as triggerable
OPEN	Alert is active and visible
ACTIVE	Ongoing trigger condition after creation
ACKNOWLEDGED	Human has explicitly received/accepted the alert; condition may still be active
SUPPRESSED	Temporarily blocked by policy/context/cooldown
RESOLVED	Trigger condition has cleared or resolution rule satisfied

VIII.2 Alert types

Type	Typical trigger basis	Notes
WASTE	Sustained qualifying waste above configured cost/duration conditions	Primary operational alert.
OFFLINE	Node/gateway unavailable beyond freshness threshold	Must distinguish device communication loss from machine OFF.
DRIFT	ACTIVE condition anomaly persists beyond condition threshold	Severity depends on calibrated condition policy.

VIII.3 Alert rule primitives

Primitive	Contract
on-delay	Condition must persist for minimum duration before opening
off-delay	Condition must remain clear for minimum duration before resolving
deadband	Creates separation between trigger and clear thresholds where numeric values are used
dedupe key	Stable identity, typically machine_id + alert type + relevant subtype
cooldown	Minimum time before re-alerting after clear/retrigger

Primitive	Contract
suppression	Explicit temporary block with reason and expiry
escalation	Change audience/severity after duration or repeated recurrence
acknowledgement	Human workflow state; must not equal resolution

VIII.4 Anti-spam and burden monitoring

Industrial alarm-management practice emphasizes alarm burden, nuisance reduction and rationalized thresholds. VOLTIX should monitor alerts/minute, repeated-alert rate, acknowledgement delay, stale-open duration, false-alarm rate, missed-alarm rate and detection delay. Any external alarm benchmark is treated as a reference for design, not a universal VOLTIX acceptance threshold.

VIII.5 Alert contract

Field	Meaning
alert_id	Stable lifecycle identity
machine_id / site_id	Scope
type	WASTE / OFFLINE / DRIFT
severity	LOW / MEDIUM / HIGH / CRITICAL according to product policy
title/message	Explainable owner-facing text
trigger_snapshot	Evidence that caused creation
opened_at / resolved_at	Lifecycle timestamps
dedupe_key	Suppression of duplicates
cooldown_until	Re-notification gate
acknowledged_by / acknowledged_at	Human acknowledgement
rule_version	Alert logic provenance
resolution_reason	Why alert ended

PART IX | LAYER 6 - ACTION / AUTOCUT

Purpose. Defines safe actuation authority, eligibility, command lifecycle, local revalidation, physical feedback and recovery behaviour.

IX.1 Action modes

Mode	Meaning
MONITOR	System observes and proves only; no notification or actuation obligation.
ALERT	System may notify; no automatic actuation.
AUTO	System may issue approved automatic actions when every safety/policy gate is satisfied.

IX.2 Never-AUTO starter list

- CNCs / machining centres mid-cycle
- Furnaces / process heat
- Unknown or unclassified machines
- Safety-critical auxiliaries
- Loads above installed actuation rating
- Any machine not explicitly whitelisted
- Loads with failed override or interlock validation

IX.3 AutoCut eligibility AND-gate

Gate family	Required condition
Evidence	Canonical Pulse state valid; sustained qualifying waste; energy residual valid; freshness within action window
Confidence	Classification meets configured AutoCut confidence policy; uncertain/unavailable blocks action
Policy	Machine/site mode permits AUTO; explicit whitelist; approval active and not expired
Safety	Not never-AUTO; not in maintenance/LOTO; no active manual override; safety classification permits action
Hardware	Actuator installed, rated, healthy and associated with correct machine
Interlock	Any required interlock/feedback preconditions satisfied
Local authority	Edge/gateway has authoritative local configuration and recent policy state
Revalidation	Immediately before dispatch, local state still satisfies the action contract

IX.4 Action sequence

1. Qualify candidate.
2. Resolve policy permission.
3. Check local hardware/interlock/override conditions.
4. Revalidate evidence immediately before dispatch.
5. Authorize and sign command.
6. Dispatch command with expiry and idempotency identity.
7. Verify physical outcome where feedback exists.
8. Record success, denial, timeout or failure.
9. Lock out or degrade further automatic actions after configured failure conditions.

IX.5 Command vs outcome

An ActionCommand is not proof that power was actually removed. The system must distinguish REQUESTED, ACCEPTED, EXECUTED, VERIFIED, FAILED, EXPIRED and DENIED semantics. Where physical feedback is not available, the product must not claim verified actuation; it may report command acceptance only.

IX.6 Fail-safe

Safety invariant Cloud loss, broker loss, ambiguous classification, stale policy, command expiry, failed preconditions or local uncertainty must not cause an unintended cut. The safe default for the production load is power ON.

IX.7 Restore behaviour

V1 does not automatically restore a machine solely because a timer elapsed or a later condition changed. Restoration should require explicit workflow/policy unless a formally approved recovery mode exists. Manual override restores power and latches the system against repeated automatic action until the override state is cleared according to policy.

IX.8 Maintenance and LOTO

AutoCut is not a substitute for electrical isolation, lockout/tagout, maintenance procedure or licensed electrical practice. Policy must allow maintenance/LOTO to suppress automatic actions and make the suppression visible in the decision trace.

IX.9 Formal Action state machine

State	Meaning	Legal next states
CANDIDATE	Waste/condition evidence suggests a possible action opportunity.	QUALIFIED, INELIGIBLE, DEFERRED
QUALIFIED	Evidence and temporal gates pass; permission still unresolved.	ELIGIBLE, BLOCKED, DENIED, SUPPRESSED
ELIGIBLE	All evidence, policy, safety, hardware and local gates pass.	AUTHORIZED, BLOCKED, DENIED, EXPIRED
AUTHORIZED	Command has explicit authorization and provenance but has not yet been dispatched.	DISPATCHED, EXPIRED, REVOKED
DISPATCHED	Authenticated command was sent to target execution path.	ACCEPTED, FAILED, EXPIRED
ACCEPTED	Target acknowledged receipt/acceptance.	EXECUTED, FAILED
EXECUTED	Requested physical/control operation is reported as executed.	VERIFIED, FAILED
VERIFIED	Physical/operational outcome met the verification contract.	TERMINAL
FAILED	Execution or verification failed.	LOCKED_OUT, DEFERRED, HUMAN_REVIEW
DENIED / BLOCKED / EXPIRED / REVOKED	Action cannot proceed under current authority/evidence.	REQUALIFY

Action state rule Decision, authorization, command dispatch and physical outcome are distinct state transitions. No UI or downstream service may infer VERIFIED from ACCEPTED or EXECUTED unless the applicable verification contract is satisfied.

IX.10 Command security and replay semantics

Control	Contract
Command identity	Every command has a unique command_id and target binding.
Expiry	Commands are valid only within an explicit expiry window; expired commands are ignored.
Authentication	Commands require authenticated/authorized origin and target verification.
Idempotency	Repeated delivery of the same command_id must not create repeated physical actions.
Replay protection	Previously executed/expired command identities cannot be re-executed.
Revocation	A revoked or superseded authorization prevents dispatch even if an old command remains otherwise valid.
Outcome verification	Feedback determines VERIFIED; absence of feedback limits claims accordingly.

PART X | LAYER 7 - M&V

Purpose. Defines evidence-grade savings reporting and prevents the platform from turning a before/after difference into an unqualified savings claim.

X.1 M&V model

Savings are conceptually the adjusted baseline energy minus reporting-period energy, with routine/non-routine adjustments and documented exclusions where applicable. The product must preserve whether an output is ESTIMATED or VERIFIED.

X.2 Required periods and evidence

Component	Contract
Baseline period	Defined period representing pre-intervention/reference performance
Reporting period	Defined post-intervention/measurement period
Operating conditions	Production, schedule, occupancy/load context as available
Adjustments	Routine and non-routine adjustments explicitly represented where material
Intervention record	When and what change/action occurred
Verification	Operational verification that the expected intervention effect occurred
Uncertainty	Known measurement/estimation limitations represented
Claim status	ESTIMATED or VERIFIED, never implicit

X.3 M&V outputs

Output	Meaning
baseline_energy	Reference energy over baseline period
reporting_energy	Measured/estimated energy over reporting period
adjusted_baseline	Baseline normalized to reporting conditions
delta_energy	Adjusted baseline minus reporting energy
estimated_savings	Decision-support savings estimate
verified_savings	Savings supported by the applicable verification procedure
confidence/uncertainty	Evidence quality/uncertainty descriptor
assumptions	Voltage, PF, tariff, production or schedule assumptions
baseline_version	Baseline provenance
report_version	M&V logic provenance

X.4 Baseline invalidation/reset

- Major process change
- Machine retrofit affecting load profile
- Substantial production regime change
- Sensor replacement or binding correction
- Extended data-quality corruption
- Known maintenance or equipment replacement

When baseline validity is broken, the system must mark the affected window and define whether a new baseline is required rather than silently comparing non-comparable periods.

X.5 Evidence export

V1 exports should support CSV and PDF-style evidence reports containing period definitions, baseline method, reporting method, intervention notes, estimates, assumptions, data-quality exclusions, and verification status. Presentation must never omit the estimation qualifier merely because the value is visually polished.

PART XI | TEMPORAL & CONTEXT

Purpose. Separates when something happened from the operating context in which it happened, and makes timing/context available without granting hidden authority.

XI.1 Temporal primitives

Primitive	Use
schedule	Expected operating plan
shift	Human/production shift context
planned_downtime	Known non-production interval
maintenance_window	Known intervention/maintenance interval
context_snapshot	Resolved context at decision time
freshness_age	Age of evidence

Primitive	Use
duration	Elapsed state/condition/alert/action duration

XI.2 Context semantics

Time answers “when?” Context answers “under what conditions?” The absence of context is UNKNOWN, not FALSE. Context can inform prioritization, alerting or qualification; authority remains in the Policy and Safety layers.

XI.3 Conflict handling

Conflict	Required behaviour
Schedule says OFF, Pulse says ACTIVE	Do not silently override Pulse; expose context conflict and apply policy-specific interpretation.
Maintenance says active lockout, waste logic says candidate	Maintenance/policy deny takes precedence for action.
Offline status and machine OFF indistinguishable from telemetry	Classify as communication uncertainty, not machine OFF.
Old policy vs newly effective policy	Use effective-time semantics and local authority rules; trace the policy actually used.

PART XII | POLICY

Purpose. Defines explicit authority and permission. Policy is not hidden inside individual engines.

XII.1 Policy hierarchy

Layer	Typical authority
Safety invariants	Non-overridable product safety constraints
Site policy	Site-specific operating rules
Machine policy	Machine classification and approval
User role	Who may approve/override/configure
Temporary override	Time-bounded operational exception
Engine recommendation	Evidence-driven suggestion only

XII.2 Deny precedence

Authority rule Deny > Allow. A single explicit deny at a higher-authority layer blocks action. “Not specified” does not equal allow.

XII.3 Policy lifecycle

State	Meaning
DRAFT	Created but not active
VALIDATED	Schema/constraint validation passed
APPROVED	Authorized for activation

State	Meaning
ACTIVE	Effective for decisions
EXPIRED	Past explicit expiry
REVOKED	Manually withdrawn
SUPERSEDED	Replaced by newer effective policy

XII.4 Local vs cloud authority

Cloud is the system of record for long-term administration where appropriate, but safety-relevant local policy and revalidation must be available at the edge. Cloud loss must not create an implicit permission to actuate.

XII.5 Policy conflict resolution

1. Apply hard safety prohibitions.
2. Apply maintenance/override constraints.
3. Apply site and machine deny/allow policy according to precedence.
4. Check approval/expiry and effective time.
5. Verify hardware and local state.
6. Only then permit action dispatch.

PART XIII | CROSS-LAYER RULES

Purpose. Defines the dependency graph and prevents semantic duplication, circular authority or hidden re-implementation.

XIII.1 Dependency graph

Canonical flow Ingest/DQ -> Pulse -> Energy -> Condition -> Ranker -> Alerts -> Action -> M&V;
Temporal/Context and Policy provide cross-cutting context. Apps consume canonical outputs and do not redefine business logic.

XIII.2 Evidence flows downward; authority does not

Pulse may provide state evidence to Energy and Ranker. Energy may provide cost evidence to Ranker and Alerts. Condition may provide drift annotation. None of those evidence layers silently authorize actuation. Authorization belongs to Policy + Action/Safety.

XIII.3 Canonical output consumption

- Downstream layers consume published canonical upstream outputs.
- If an upstream value is invalid/stale/uncertain, the downstream rule must respect that status.
- A downstream consumer may derive a new metric from an upstream output, but must not redefine its semantic meaning.
- Any exception must be documented as a new contract or rule, not hidden in implementation.

XIII.4 Cross-layer conflict matrix

Conflict	Resolution
Pulse uncertain vs waste rule trigger	No decision-grade waste qualification; human review may show the signal.

Conflict	Resolution
Energy baseline invalid vs ranker	Exclude from monetary ranking or show degraded/non-qualified state.
Condition drift vs waste	Show independently; neither automatically creates the other.
Ranker says “act” vs Policy deny	Deny wins; recommendation remains a suggestion.
Alert open vs Action not eligible	Alert may remain open; action eligibility is independently evaluated.
Action command accepted vs physical outcome unknown	Do not claim execution verified.
M&V savings estimate vs insufficient verification	Mark ESTIMATED; do not relabel VERIFIED.

XIII.5 Decision trace continuity

Every material cross-layer outcome should be traceable from the final decision to the upstream evidence snapshots and versions that justified it. This supports debugging, audits, replay and stakeholder trust.

XIII.6 Architecture dependency graph

Dependency rule The following is a dependency map, not a single linear pipeline. Shared Context, Temporal and Policy provide cross-layer inputs; downstream consumers use canonical outputs and retain their own authority boundaries.

Consumer	Primary dependencies	Must not reimplement
Pulse	DQ + telemetry + machine configuration + temporal freshness	Raw transport validation already owned by DQ; downstream state definitions.
Energy	DQ + Pulse + baseline + tariff + voltage/PF provenance	Pulse state interpretation.
Condition	DQ + Pulse ACTIVE + healthy reference + operating regime + model state	Pulse or Energy semantics.
Ranker	Pulse + Energy + optional Condition + Policy visibility + temporal context	New Pulse/Condition classifications.
Alerts	Canonical layer outputs + Temporal + Policy + existing alert lifecycle	Independent duplicate ranking/classification logic.
Action	Canonical evidence + Policy + Safety + hardware + override + local context	Trusting an upstream recommendation as authorization.
M&V	Energy + intervention/action records + temporal/context + baseline	Calling command execution proof by assumption.
Apps	Canonical outputs + explanations + statuses	Business-rule decisions hidden in frontend presentation code.

XIII.7 Authority classes

Authority class	Typical examples	Change control expectation
PHYSICAL	Electrical power relationships, measured hardware capability	Engineering verification; cannot be overridden by product preference.
STATISTICAL	Model score interpretation, calibration	Model owner + validation evidence.

Authority class	Typical examples	Change control expectation
	and readiness	
SAFETY	Never-AUTO, fail-safe ON, override, interlock requirements	Explicit safety review/approval.
POLICY	Mode, whitelist, approval, deny/allow, expiry	Authorized policy owner.
SITE	Tariff, schedule, operating constraints, maintenance windows	Site-authorized configuration.
MACHINE	Machine class, baseline, regime, capability mapping	Asset/engineering owner.
PRODUCT	Rank ordering, user-facing alert behaviour	Product/rules owner with QA evidence.
BUSINESS	Economic prioritization and reporting interpretation	Business/product approval; cannot override safety.
CONFIGURATION	Bindings, versions, deployment parameters	Controlled configuration lifecycle.

XIII.8 Model evaluation modes

Mode	Decision produced	Production side effect
LIVE	Yes	May alert or act only when all production gates pass.
SHADOW	Yes, isolated/logged	No alerting or physical actuation side effect.
DRY_RUN	Yes, historical/simulated	No production side effect; suitable for replay and rule/model testing.

XIII.9 Model lifecycle

State	Meaning
NOT_READY	Required training/validation evidence does not exist.
TRAINING	Model artefact under construction; never used for production decisions.
SHADOW	Model runs against live or replayed data without production authority.
VALIDATION	Formal acceptance evidence is being evaluated.
APPROVED	Meets the defined promotion criteria but not yet active.
ACTIVE	Current production model/version.
STALE	No longer trusted for current decision scope.
SUPERSEDED	Replaced by a newer approved model.
RETIRED	Permanently withdrawn from use; historical decisions retain provenance.

XIII.10 Configuration lifecycle

State	Meaning
DRAFT	Configuration proposed but not validated.
VALIDATED	Configuration passes structural/semantic checks.
ACTIVE	Configuration is currently effective.
SUPERSEDED	Replaced by newer configuration; historical decisions retain old version.
REVOKED	Explicitly invalidated before natural replacement/expiry.

PART XIV | APPLICATIONS / UX RULES

Purpose. Defines how semantic states and uncertainty must appear to humans without moving business logic into frontend code.

XIV.1 UX semantic principles

- Rupees before vanity charts.
- Ranked problems before generic dashboards.
- Waste and condition are visually distinct concepts.
- Estimated, stale, uncertain and unavailable states remain visible.
- Never-AUTO and fail-safe status are discoverable.
- Commanded vs physically verified action outcomes are distinct.
- OLED, web and mobile surfaces should preserve the same semantic labels.

XIV.2 Core screens

Screen	Required semantic content
Ranked waste	Machine, current INR/hr, duration, state, explanation, evidence freshness, action annotation
Machine detail	Pulse history, residual, baseline status, condition drift, alerts, action history
Alerts	Lifecycle state, severity, reason, acknowledgement, clear condition
AutoCut approval	Mode, whitelist, safety status, deny reason, approval expiry, override state
M&V	Baseline/reporting periods, adjusted baseline, estimates, assumptions, verification status
Node health	Freshness, connectivity, firmware, buffering, binding/commissioning status
Commissioning	CT correctness, OFF/ACTIVE/IDLE captures, tariff, override test, never-AUTO classification

XIV.3 Frontend ownership boundary

No frontend-owned business logic UI components may format and filter canonical outputs, but they must not

invent ranking formulas, alert thresholds, action eligibility, state semantics or M&V calculations.

PART XV | RULE REGISTRY

Purpose. Defines the formal registry and lifecycle for every rules artefact used by VOLTIX.

XV.1 Registry fields

Field	Meaning
rule_id	Stable semantic identifier
domain	DQ / PLS / ENG / COND / RANK / ALT / ACT / MV / TMP / POL / CFG etc.
rule_type	Interpretation / Validation / Quantification / Qualification / Decision / Policy / Temporal / Safety / Presentation / Proof
purpose	One-sentence business purpose
inputs	Canonical upstream fields/statuses
conditions	Logical qualification
output	Decision/result semantics
failure_mode	What happens when evidence is insufficient
owner	Responsible engineering/product owner
version	Immutable rule version
lifecycle_state	DRAFT/TESTING/APPROVED/ACTIVE/SUPERSEDED/RETIRED
effective_from/to	Time validity
parameter_refs	Threshold/calibration dependencies
model_refs	Model dependencies if any
audit_required	Yes/No
source_basis	Document/research/decision source

XV.2 Rule lifecycle

State	Gate
DRAFT	Designed but not executable
TESTING	Acceptance/boundary/failure tests running
APPROVED	Review sign-off complete
ACTIVE	Effective for production decisions
SUPERSEDED	Replaced but retained for historical trace
RETIRED	No longer valid and no longer executable

XV.3 Initial rule namespace

Namespace	Scope
DQ-*	Data quality / ingest
PLS-*	Pulse / state semantics
ENG-*	Energy / residual / economics
COND-*	Condition / drift
RANK-*	Waste ranker
ALT-*	Alerts
ACT-*	Action / AutoCut
MV-*	M&V / proof
TMP-*	Temporal/context
POL-*	Policy
CFG-*	Configuration/commissioning

XV.3 V1 rule inventory

This inventory is the minimum semantic map. Detailed logic may evolve within the contract, but an identified rule responsibility must have a stable owner, status and registry entry rather than existing as undocumented code behaviour.

Domain	Rule IDs / responsibility family	Primary responsibility
DQ	DQ-001..DQ-012	Admission, integrity, freshness, sequence, buffering and recovery.
COM	COM-001..COM-012	Commissioning, binding, capture, safety and readiness admission.
PLS	PLS-001..PLS-0xx	Canonical Pulse interpretation, debounce, hysteresis, confidence and model boundary.
ENG	ENG-001..ENG-0xx	Power estimate provenance, baseline, residual, energy accumulation and economics.
COND	COND-001..COND-0xx	ACTIVE-only condition feature quality, regime, anomaly score, persistence and model readiness.
RANK	RANK-001..RANK-0xx	Candidate qualification, cost-rate ordering, tie-breaks, significance and explainability.
ALT	ALT-001..ALT-0xx	Alert trigger, delay, lifecycle, suppression, dedupe, escalation and burden metrics.
ACT	ACT-001..ACT-0xx	Action eligibility, authorization, command lifecycle, local recheck, feedback and lockout.
MV	MV-001..MV-0xx	Baseline validity, normalization, intervention evidence, verification and

Domain	Rule IDs / responsibility family	Primary responsibility
		claim state.
TMP	TMP-001..TMP-0xx	Time, duration, freshness, schedules, persistence, expiry and temporal conflict.
POL	POL-001..POL-0xx	Authority, deny precedence, approval, expiry, revocation and overrides.
CFG	CFG-001..CFG-0xx	Configuration validity, effective version and deployment-state management.
UX	UX-001..UX-0xx	Presentation semantics: stale/estimated/uncertain/blocked/verified labels; no frontend business logic.

XV.4 Rule lifecycle and approval

Lifecycle	Meaning	Promotion requirement
DRAFT	Being defined; not production-authoritative.	Semantic review.
TESTING	Implemented or simulated against acceptance cases.	Test evidence recorded.
APPROVED	Accepted for production activation.	Owner + required approvers sign off.
ACTIVE	Current effective rule.	Effective date/version recorded.
SUPERSEDED	Replaced by newer rule.	Historical provenance retained.
RETIRED	No longer valid for new evaluation.	Removal/archival reason recorded.

XV.5 Rule registry authority fields

Field	Required purpose
authority_class	Identifies which source of authority governs the rule.
change_authority	Role/person/team that may approve semantic changes.
effective_scope	Site, machine class, deployment or product scope.
precedence	Priority relative to conflicting rules.
evaluation_mode	LIVE / SHADOW / DRY_RUN availability.
safety_impact	Whether a failure could affect physical safety/production continuity.

PART XVI | PARAMETER REGISTRY

Purpose. Separates adjustable thresholds from rule meaning and preserves their basis, ownership and calibration state.

XVI.1 Parameter registry fields

Field	Meaning
parameter_id	Stable identifier
value + units	Configured value
scope	Global/site/machine/model
source	Default, commissioning, calibration, policy, research
basis	Evidence or rationale
owner	Responsible approver
effective_version	Parameter version
calibration_state	UNCALIBRATED / CALIBRATED / VALIDATED
effective_from/to	Temporal validity
safety_critical	Whether extra approval controls apply
change_reason	Why it changed

XVI.2 Example parameter families

Family	Examples
Pulse	state thresholds, debounce, hysteresis, confidence policy
Energy	baseline window, residual smoothing, tariff mapping
Condition	feature windows, persistence, calibrated anomaly threshold
Ranker	minimum duration, ranking hysteresis, tie-break ordering
Alerts	on-delay, off-delay, cooldown, severity thresholds
Action	minimum waste duration, confidence gate, command expiry, lockout duration
M&V	period definitions, normalization factors, quality gates
Temporal	freshness limits, schedule tolerance

XVI.3 Numeric values and scientific claims

No single numeric value is treated as “scientifically proven for all machines” without evidence. Where an empirical value is needed, the registry must identify its calibration method, source, deployment scope and validation criteria. Machine/site calibration is preferred for thresholds that depend strongly on operating behaviour.

XVI.3 V1 parameter inventory

Parameter ID	Family	Unit / semantic type	Owner	Basis / status
PLS_DEBOUNCE_DURATION	Pulse	duration	Pulse owner	Commissioning/model validation; calibrate.
PLS_CONFIDENCE_GATE	Pulse	dimensionless threshold	Pulse + safety owner	Machine/policy specific; no universal scientific

Parameter ID	Family	Unit / semantic type	Owner	Basis / status
				default.
PLS_STATE_HYSTERESIS	Pulse	duration/threshold set	Pulse owner	Validated state-transition behaviour.
ENG_BASELINE_WINDOW	Energy	time window	Energy owner	Machine behaviour/history.
ENG_RESIDUAL_MIN	Energy	kW / threshold	Energy owner	Noise and measurement quality.
ENG_TARIFF_RATE	Economics	currency/kWh	Site owner	Effective-dated site tariff/provenance.
COND_PERSISTENCE_DURATION	Condition	duration	Condition owner	Healthy validation and field review.
COND_DECISION_THRESHOLD	Condition	score/threshold	Condition owner	Calibrated against validation distribution.
RANK_MIN_QUALIFYING_DURATION	Ranker	duration	Rules owner	Operational usability and waste behaviour.
RANK_COST_RATE_FLOOR	Ranker	currency/hour	Rules owner	Minimum economically meaningful rate.
ALT_WASTE_ON_DELAY	Alerts	duration	Rules owner	Nuisance/burden validation.
ALT_OFF_DELAY	Alerts	duration	Rules owner	Stable resolution semantics.
ALT_COOLDOWN	Alerts	duration	Rules owner	Anti-spam evidence.
ACT_EVIDENCE_FRESHNESS	Action	duration	Safety/action owner	Machine/action specific validation.
ACT_COMMAND_EXPIRY	Action	duration	Safety/action owner	Replay/failsafe design.
ACT_FAILURE_LOCKOUT	Action	count/time	Safety/action owner	Physical outcome failure response.
TMP_PULSE_FRESHNESS	Temporal	duration	Platform/rules owner	Layer-specific freshness contract.
MV_BASELINE_MIN_VALIDITY	M&V	quality/status rule	M&V owner	Project methodology.

XVI.4 Parameter provenance and change control

A parameter is not permitted to become an unexplained magic number in implementation. Every parameter must identify source/basis, unit, owner, approval status, effective version, scope and calibration state. Changing a parameter version must be auditable even when the rule version is unchanged.

PART XVII | FAILURE / DEGRADED / RECOVERY BEHAVIOR

Purpose. Defines how VOLTIX behaves when evidence, connectivity, models, policy, baselines or actuators fail.

XVII.1 Failure matrix

Failure	Detection	Degraded mode	Recovery requirement
CT/sensor invalid	Range/completeness/health check	No trusted residual; rank/action gates fail closed	Revalidate sensor and commissioning
Node offline	Freshness/heartbeat timeout	History retained; no new local decisions that require missing evidence	Heartbeat recovery and state reconciliation
Gateway offline	Local health	Local safety operation continues; buffer locally if possible	Reconnect and reconcile buffered history
Cloud unavailable	Connectivity failure	No loss of safety function; local operation continues	Resync policy/config and upload history
Pulse model unavailable	Model health	Use deterministic rules if contract permits, else classification unavailable	Model readiness restored and version traced
Condition model stale	Model age/validation state	Condition flags downgraded/unavailable	Recalibrate/redeploy model
Baseline invalid	Quality/invalidation rule	No qualified monetary waste or M&V claim	Rebuild/revalidate baseline
Policy stale/unknown	Version/freshness check	No new AutoCut	Sync validated policy or remain safe
Override active	Override state	Force/maintain power ON; block AutoCut	Explicit override clear + requalification
Command expired	Expiry timestamp	No actuation	Issue new authorized command if still needed
Actuator failure	Feedback mismatch/health	Action marked failed; automatic retry subject to lockout policy	Human/service intervention and re-test
Config mismatch	Version/binding check	Block affected decisions/actions	Commissioning/config reconciliation

XVII.2 Recovery principles

- Recovery is not equivalent to backfilling assumptions.
- A recovered connection does not retroactively execute an expired command.
- A recovered sensor must pass the same validation as a newly admitted sensor.
- Model recovery must preserve version provenance; historical decisions remain tied to historical versions.
- Automatic re-entry into AUTO requires explicit requalification where configured.

PART XVIII | AUDIT & DECISION TRACE

Purpose. Creates the liability and trust spine for recommendations, policy changes, overrides and actuation.

XVIII.1 Append-only audit events

Event family	Minimum content
AutoCut decision	Machine, decision, reason, evidence snapshot, rule/policy/model

Event family	Minimum content
	versions, evaluator, timestamp
Command	Command id, action, target, policy version, expiry, signature/auth context
Override	Who, when, scope, reason, expiry/clear action
Mode change	Old/new mode, authority, reason, effective time
Policy change	Old/new policy versions, approver, effective time
Baseline reset	Trigger, old/new baseline, operator/model provenance
Alert lifecycle	Open/ack/suppress/resolve events and reasons
Commissioning	Tests performed, pass/fail, operator, hardware/config versions

XVIII.2 Decision trace

A decision trace should reconstruct: input context -> validation -> canonical upstream evidence -> rule evaluation -> policy resolution -> final decision -> action command (if any) -> physical outcome -> audit event. The trace must retain enough historical snapshots to avoid dependence on today's values for explaining yesterday's decision.

XVIII.3 Replay

Historical replay is for engineering analysis and audit. Replay must not have side effects on physical actuation. Safety-sensitive code paths should expose a dry-run/shadow evaluation mode so historical or candidate rule changes can be tested without dispatching commands.

PART XIX | TEST / QA SPECIFICATION

Purpose. Defines the acceptance gates required before a rule, model, policy or action capability is considered production-ready.

XIX.1 QA categories

Category	Examples
Functional	Nominal state, residual, rank, alert, action, M&V flows
Boundary	Threshold equality, minimum duration, zero/negative residual, confidence floor
Temporal	On-delay/off-delay, gaps, out-of-order, late events, expiry
Data quality	Duplicates, corrupt values, stale data, unbound machines
Failure	Offline, broker loss, cloud loss, model unavailable, actuator failure
Safety	Never-AUTO, override, policy deny, stale policy, replayed command
Adversarial	Spoofed device, unauthorized command, forged sequence, duplicate action
Model	Shadow vs active agreement, calibration, contaminated training data
Regression	Rule/version change does not break historical semantic contracts

Category	Examples
UX semantics	Estimated/stale/uncertain labels are never dropped

XIX.2 Acceptance checklist

- Every rule has explicit inputs, output, failure mode, owner and version.
- Every safety decision has a deny path and an audit event.
- All downstream engines consume canonical upstream fields/statuses.
- No action can be dispatched from stale/uncertain/unavailable required evidence.
- Offline/cloud-loss behaviour has been tested for safe continuity.
- Action commands are authenticated, expiring and idempotent.
- Physical action outcome is distinguished from command acceptance.
- M&V exports preserve assumptions and verification status.
- Model promotion and baseline reset paths are tested before activation.

XIX.3 Rule acceptance table template

Field	Example content
Given	Machine is ACTIVE then enters IDLE
When	IDLE persists for configured minimum and residual is valid
Then	Candidate qualifies for waste ranking
And	AutoCut remains false unless policy, whitelist, safety and local recheck all pass
Reason	Sustained avoidable powered nonproductive operation
Evidence	Pulse/energy snapshot and versions recorded
Failure	Uncertain Pulse -> no qualification

XIX.4 Safety/adversarial tests

- Replay a previously valid CUT command after expiry.
- Send CUT while override is active.
- Send CUT for never-AUTO machine.
- Corrupt policy version or make it stale.
- Break gateway/cloud link immediately before dispatch.
- Send duplicate command IDs.
- Make physical feedback disagree with command result.
- Inject out-of-order events that would appear to prolong waste.

PART XX | WORKED EXAMPLES

Purpose. Shows end-to-end semantic traces across multiple layers. Examples are illustrative, not universal numeric defaults.

XX.1 Example A - Compressor sustained waste

Stage	Outcome
Ingest	CT telemetry valid; machine bound; freshness valid.

Stage	Outcome
Pulse	State becomes IDLE after debounce; classification_status VALID.
Energy	Estimated power 5.8 kW; baseline necessary 1.2 kW; avoidable 4.6 kW; tariff makes current rate INR 52/hr.
Ranker	Qualifies and rises to top by current avoidable cost rate.
Alert	WASTE alert opens after configured on-delay; dedupe key compressor-2 + WASTE.
Policy	Machine is AUTO-approved and not in maintenance.
Action	Local recheck passes; command expires after short validity window; command dispatched.
Outcome	Feedback confirms physical OFF; trace records VERIFIED execution.
M&V	Reporting evidence includes intervention timestamp and later normalized comparison.

XX.2 Example B - CNC drift but no AutoCut

Stage	Outcome
Ingest	Telemetry valid.
Pulse	ACTIVE, valid.
Energy	No sustained avoidable idle waste.
Condition	Electrical RMS distribution drifts from healthy centroid; drift flag persists.
Ranker	Condition annotation increases visibility but does not create monetary waste.
Alert	DRIFT alert opens with advisory severity.
Policy/Action	Machine class is never-AUTO mid-cycle; no automatic command can be issued.
UX	Operator sees condition alert and “AutoCut unavailable for this machine class.”

XX.3 Example C - Cloud outage during candidate waste

Stage	Outcome
Edge	Machine enters qualifying waste state; local evidence remains available.
Cloud	Unavailable; no cloud decision required for safety.
Action	If local policy and safety contract independently authorize, local path may act; otherwise load remains ON.
Buffering	Telemetry and decision metadata buffered for later reconciliation.
Recovery	Cloud reconnects; history is uploaded; no expired cloud

Stage	Outcome
	command is replayed.

XX.4 Example D - Baseline invalidation after process change

Stage	Outcome
Context	Machine process changes materially after retrofit.
Detection	Baseline quality rule identifies non-comparability.
Energy	Residual remains ESTIMATED/DEGRADED; monetary rank may be suppressed.
M&V	Baseline marked invalid for affected scope.
Recovery	New operating regime/reference captured; baseline version increments after validation.

XX.5 Example E - Manual override during otherwise eligible AutoCut

Stage	Outcome
Evidence	Machine enters sustained qualifying waste; Pulse/energy evidence is valid and fresh.
Qualification	Action candidate becomes QUALIFIED and then ELIGIBLE on all non-override gates.
Override	Operator activates manual override before dispatch.
Policy/Safety	Override has higher authority than the automatic action; AutoCut becomes BLOCKED.
Execution	No CUT command is dispatched; load remains ON.
Trace	Override actor, scope, reason, time and blocked decision are recorded.
Recovery	After explicit override clear, machine must be requalified; the old candidate is not automatically replayed.

XX.6 Example F - Stale/conflicting evidence

Stage	Outcome
Pulse	Last valid state is IDLE, but evidence age exceeds the action freshness window.
Context	Schedule may suggest non-production, but schedule does not prove current electrical state.
Ranker	Current cost-rate priority is marked stale/deferred rather than silently treated as current.
Action	AutoCut is BLOCKED because required evidence is stale.
UX	Owner sees the last observed state plus explicit stale timestamp/label.

Stage	Outcome
Recovery	Fresh telemetry arrives; downstream engines re-evaluate from the new context; no stale command is reused.

PART XXI | OPEN DECISIONS

Purpose. Contains only decisions that should remain explicit rather than being silently assumed by implementers.

XXI.1 Open-decision rule

Open decisions are intentionally limited to matters whose resolution materially changes implementation semantics. A developer must not close an open decision by inventing a default.

ID	Decision	Why it matters	Default posture
OD-01	Exact V1 Pulse confidence representation	Affects model contract and AutoCut qualification semantics	Keep confidence semantics model-specific; require explicit status
OD-02	Physical actuation feedback availability per hardware variant	Determines VERIFIED vs command-accepted outcome semantics	Never claim physical verification without feedback
OD-03	Final site/machine policy precedence details	Affects authorization conflicts	Deny wins; explicit precedence table must be approved
OD-04	Minimum freshness windows per layer	Affects eligibility and alert/action timing	Use parameter registry; do not use one universal window
OD-05	Final M&V methodology scope for the pilot	Affects verification and claim language	Report operational evidence with explicit limitations

PART XXII | RESEARCH & STANDARDS BASIS

Purpose. Records the standards/literature themes that inform the contract without turning external guidance into claims of universal VOLTIX values.

XXII.1 Measurement and power fundamentals

The energy layer is grounded in standard electrical power relationships and the distinction between power and energy. For single-phase systems the conceptual reference is $P = V * I * PF$; for balanced three-phase systems $P = \sqrt{3} * V_{LL} * I * PF$; energy is the time integral of power. VOLTIX may estimate these quantities from CT current plus nominal/assumed values, but the resulting provenance remains ESTIMATED unless instrumented measurements justify stronger evidence.

XXII.2 M&V basis

The M&V design follows the general structure used in IPMVP/DOE and ISO 50006/ISO 50015 practice: establish a baseline, define the reporting period, normalize/adjust for relevant conditions, document measurement boundaries and uncertainty, and distinguish operational verification from a savings claim. Exact project methodology and verification scope remain deployment decisions.

XXII.3 Condition/anomaly basis

The condition architecture is informed by statistical anomaly detection and industrial operating-regime separation. Isolation Forest and GMM-style methods are suitable implementation candidates, but score scales, contamination rates and percentile thresholds are calibration parameters, not universal constants.

XXII.4 Alarm management basis

Alert lifecycle, rationalization, nuisance suppression and burden monitoring are informed by industrial alarm-management practice such as ISA-18.2 and EEMUA-191, plus research on the trade-off among false alarms, missed events and detection delay. VOLTIX should treat any numerical benchmark as contextual guidance, not an automatic acceptance threshold.

XXII.5 Machine safety basis

Action/safety design is informed by risk-assessment and machinery-control principles reflected in ISO 12100, ISO 13849-1/-2, ISO 13850 and IEC 60204-1, together with applicable hazardous-energy/LOTO practice. VOLTIX does not claim certification merely by referencing these standards; site electrical work and compliance remain qualified professional responsibilities.

XXII.6 Research-source interpretation rule

Source	Supports	Does not support	VOLTIX interpretation
IPMVP / DOE	Baseline/reporting, normalization, verification concepts	A universal VOLTIX savings threshold	Use as M&V design basis.
ISO 50006 / ISO 50015	Energy baseline/EnPI/M&V structure	Permission to call every estimate verified	Preserve evidence status and scope.
Isolation Forest literature	Anomaly detection approach	Universal score meaning/threshold	Calibrate for machine/regime context.
GMM / operating-mode research	Regime clustering/classification	Universal number of states	Use as a model behind canonical Pulse states.
ISA-18.2 / EEMUA-191	Alarm rationalization and burden concepts	One-size-fits-all alarm thresholds	Use to shape alert lifecycle and QA metrics.
ISO 12100 / 13849 / IEC 60204	Risk/safety engineering principles	Automatic certification of VOLTIX hardware	Use as design inputs and require site-qualified compliance review.

XXII.7 References to retain with the implementation package

- VOLTIX final detailed technical/product specification (primary internal source).
- VOLTIX final quick-reference document (primary operational source).
- Rule Engine Ticket - Waste Ranker + Alerts + Product Rules Layer (formal ownership/rules source).
- IPMVP Core Concepts / DOE measurement and verification guidance.
- ISO 50006:2023 - energy management systems and energy performance indicators/baselines.
- ISO 50015:2025 - measurement and verification of energy performance within energy management systems.
- Liu, Ting and Zhou - Isolation Forest (original anomaly detection paper).
- ISA-18.2 - Management of Alarm Systems for the Process Industries.
- EEMUA-191 - Alarm Systems: A Guide to Design, Management and Procurement.
- ISO 12100 - Safety of machinery - risk assessment and risk reduction.
- ISO 13849-1 / ISO 13849-2 - safety-related parts of control systems and validation.
- ISO 13850 - emergency stop function principles.

- IEC 60204-1 - electrical equipment of machines.
- Applicable local electrical codes, hazardous-energy controls and qualified installation/testing requirements.

PART Appendices | CONTRACT DICTIONARIES & TRACEABILITY

Purpose. Provides compact reference material developers can keep beside implementation work.

A. Contract dictionary

Term	VOLTIX meaning
AutoCut	Policy-authorized automatic interruption of an eligible load; not unrestricted shutdown.
Baseline	Reference necessary/expected energy performance against which current/reporting performance is compared.
Condition drift	Evidence that ACTIVE electrical behaviour departs from a validated healthy reference.
Decision	A semantic product outcome; not necessarily a physical action.
Evidence	Upstream observation/derived result used to justify downstream reasoning.
Fail-safe ON	Default load state when safety-relevant control conditions become unavailable/ambiguous.
Policy	Explicit authority governing what actions are permitted.
Qualify	Pass a gate that makes a situation eligible for a later decision.
Residual	Difference between estimated current power and necessary baseline power.
State	Canonical interpretation of machine operating condition: OFF/ACTIVE/IDLE/WASTE.
Verification	A procedure-supported confirmation that an expected physical/operational result occurred.

B. State dictionary

State/status	Definition	Blocks AutoCut?
OFF	Negligible machine draw	No; but not a reason to issue CUT without a supported candidate/command workflow.
ACTIVE	Productive/loaded operation	Yes if action would interrupt productive operation.
IDLE	Powered/nonproductive	Not by state alone; other gates required.
WASTE	Qualified avoidable/prolonged nonproductive state	No by state alone; policy/safety gates remain mandatory.
UNCERTAIN	Classification ambiguous	Yes
UNAVAILABLE	No usable classification evidence	Yes

C. Reason-code dictionary (starter set)

Code	Meaning
DQ_INVALID	Required telemetry failed validation
DQ_STALE	Evidence outside freshness contract
PLS_UNCERTAIN	Pulse classification uncertain
ENG_BASELINE_INVALID	Energy baseline not valid
ENG_TARIFF_MISSING	Economic rate unavailable
RANK_SUSTAINED_WASTE	Qualifying waste duration reached
RANK_HIGH_COST_RATE	High current avoidable cost rate
ALT_WASTE_OPEN	Waste trigger satisfied
ALT_OFFLINE_OPEN	Node/gateway offline condition satisfied
ALT_DRIFT_OPEN	Condition drift persistence satisfied
ACT_POLICY_DENY	Policy prevents automatic action
ACT_NEVER_AUTO	Machine class prohibited from AutoCut
ACT_OVERRIDE	Manual override active
ACT_STALE_EVIDENCE	Required evidence too old
ACT_LOCAL_RECHECK_FAIL	Local safety/eligibility recheck failed
ACT_COMMAND_EXPIRED	Command expiry reached
ACT_EXECUTION_UNVERIFIED	Physical outcome could not be verified
MV_BASELINE_RESET	Baseline invalidated/reset
MV_ESTIMATED	Evidence supports estimate, not verification

D. Parameter provenance matrix

Parameter family	Owner	Basis	Approval expectation
Pulse thresholds/debounce	Pulse owner	Commissioning evidence + model validation	Technical approval
Energy baseline window	Energy owner	Machine behaviour/history	Technical approval
Condition persistence/threshold	Condition owner	Healthy validation distribution + field review	Technical + product approval
Ranker minimum duration	Rules owner	Waste behaviour / operator usability	Product approval
Alert on-delay/cooldown	Rules owner	Alarm burden / nuisance testing	Product approval
AutoCut confidence/duration	Safety + action owner	Machine-specific validation	Explicit safety approval
M&V verification thresholds	M&V owner	Project methodology	M&V approval

E. Rule dependency matrix

Consumer	Canonical upstream dependencies	Authority received?
Energy	Pulse + DQ + tariff + baseline	No
Condition	Pulse ACTIVE + DQ + healthy reference + context	No
Ranker	Pulse + Energy + optional Condition + criticality	No
Alerts	Ranker/energy/pulse/condition + temporal policy	No
Action	Pulse + Energy + Policy + Safety + hardware + override + context	Yes, after local policy/safety resolution
M&V	Energy + intervention + context + baseline/reporting windows	No
Apps	Canonical outputs from all layers	No

F. Cross-layer traceability

Need	Source contract	Primary part
Trustworthy telemetry	Ingest/DQ responsibilities	III
Canonical operating state	Pulse contract	IV
Avoidable cost quantification	Energy residual/baselines	V
Electrical condition flag	ACTIVE-only condition contract	VI
Owner priority	Explainable ranker	VII
Human interrupt	Alert lifecycle	VIII
Safe actuation	Policy-gated action	IX + XII
Savings evidence	M&V baseline/reporting	X
Time/context	Temporal primitives	XI
Auditability	Append-only trace	XVIII
QA readiness	Acceptance categories	XIX

G. Failure matrix quick reference

Failure class	Primary safe response
Evidence missing/invalid	Do not infer; degrade explicitly
Classification uncertain	No AutoCut
Policy stale/unknown	No new AutoCut
Override active	Power ON / block AutoCut
Cloud lost	Local safe operation; no cloud-required cut
Command expired	Ignore; remain ON

Failure class	Primary safe response
Actuator outcome mismatch	Mark failed; lockout/review
Baseline invalid	Suppress decision-grade savings claims

H. Honest-claims matrix

Claim	V1 position
Continuous per-machine electrical observation	Yes, CT-first where installed
Operating-state intelligence	Yes
Waste prioritization	Yes
Safe, policy-gated actuation	Yes for eligible/whitelisted loads and qualified implementation
Electrical ACTIVE drift flagging	Yes
Operational M&V-style evidence	Yes
Utility billing accuracy	No
Full predictive maintenance	No
Quality yield from current alone	No
Air-leak localization without flow	No
Unrestricted AutoCut	No

I. Implementation handoff statement

A developer receiving this document should be able to determine what each layer means, which evidence it consumes, what it may decide, what it must never decide, how uncertainty propagates, which versions must be recorded, how failures degrade, and how the result is tested. The developer remains free to choose the implementation technology as long as the observable semantic contract is preserved.

PART Change Log | DOCUMENT CHANGE LOG

Version	Date	Status	Summary
1.0	2026-09-07	Baseline	Complete V1 contract specification assembled from the VOLTIX source documents and established cross-layer architecture decisions.
1.1	2026-09-07	Revision	Closed completeness-audit gaps: canonical evaluation context, orthogonal status taxonomies, decision states, commissioning lifecycle/rules, formal action state machine, dependency/authority model, evaluation modes, model/config lifecycles, expanded rule/parameter inventories, unit semantics and worked

Version	Date	Status	Summary
			examples.